Mohamed Adan Satawa

Moringa School

Part Time 14 Class of 2025

Project one, Phase One

BUSINESS UNDERSTANDING

Business Understanding

Investing in aircraft that will pose lowest safety risk during operation

Data Source was from Kaggle site where data was in Ms Excel format.

Original data set had 2,500 rows and 8 columns.

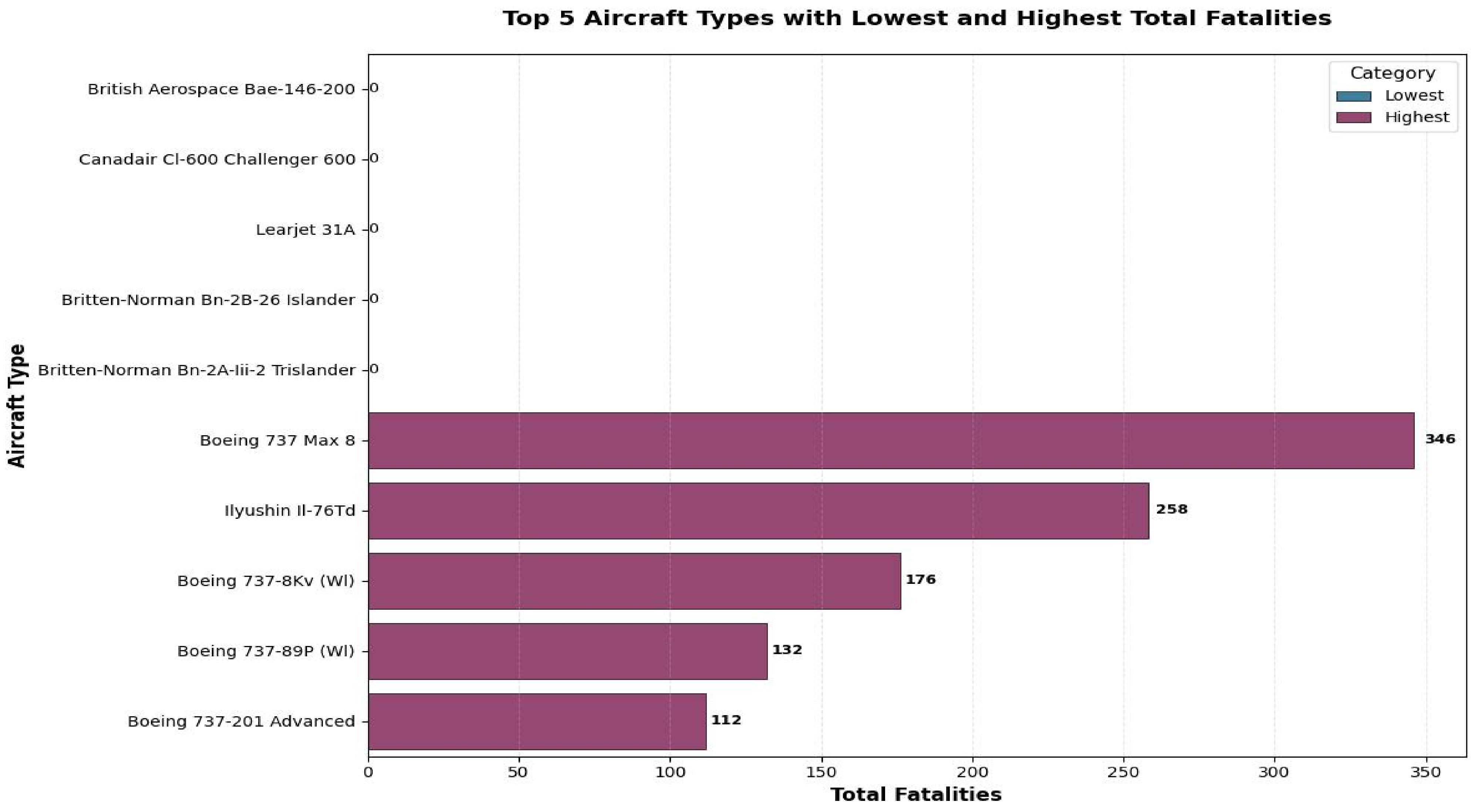
Data Analysis - Pandas library was used to do .info, .describe and .describe(include='all') was done for both numerical and categotical data to have overview of dataset descriptive

Data Visualization

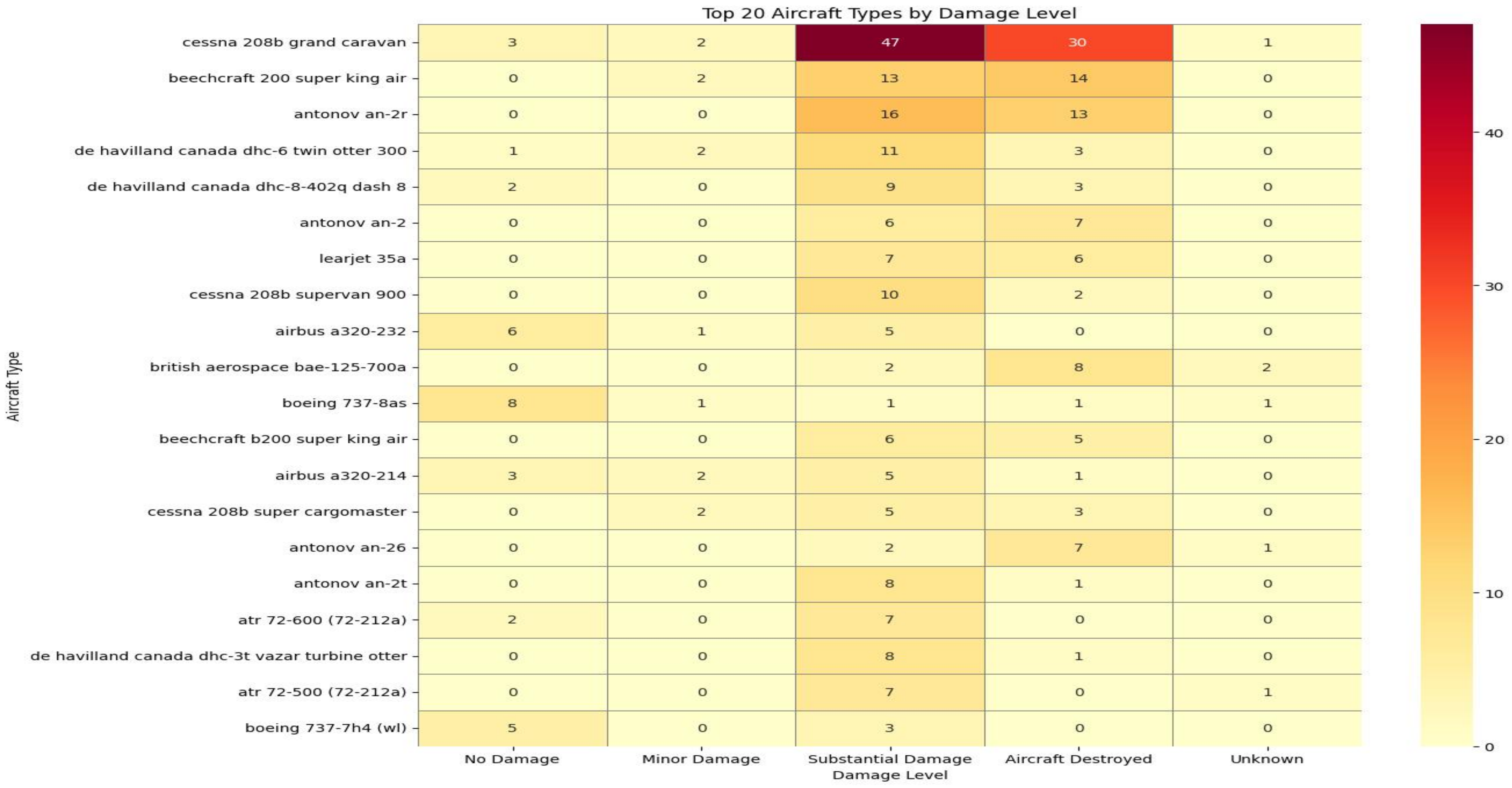
Visualization was done based on specific objectives, which are:-

- Aircraft_type with lowest and highest fatalities
- Damage level for aircraft type
- Fatalities by the month of the year
- Location with lowest and highest fatalities

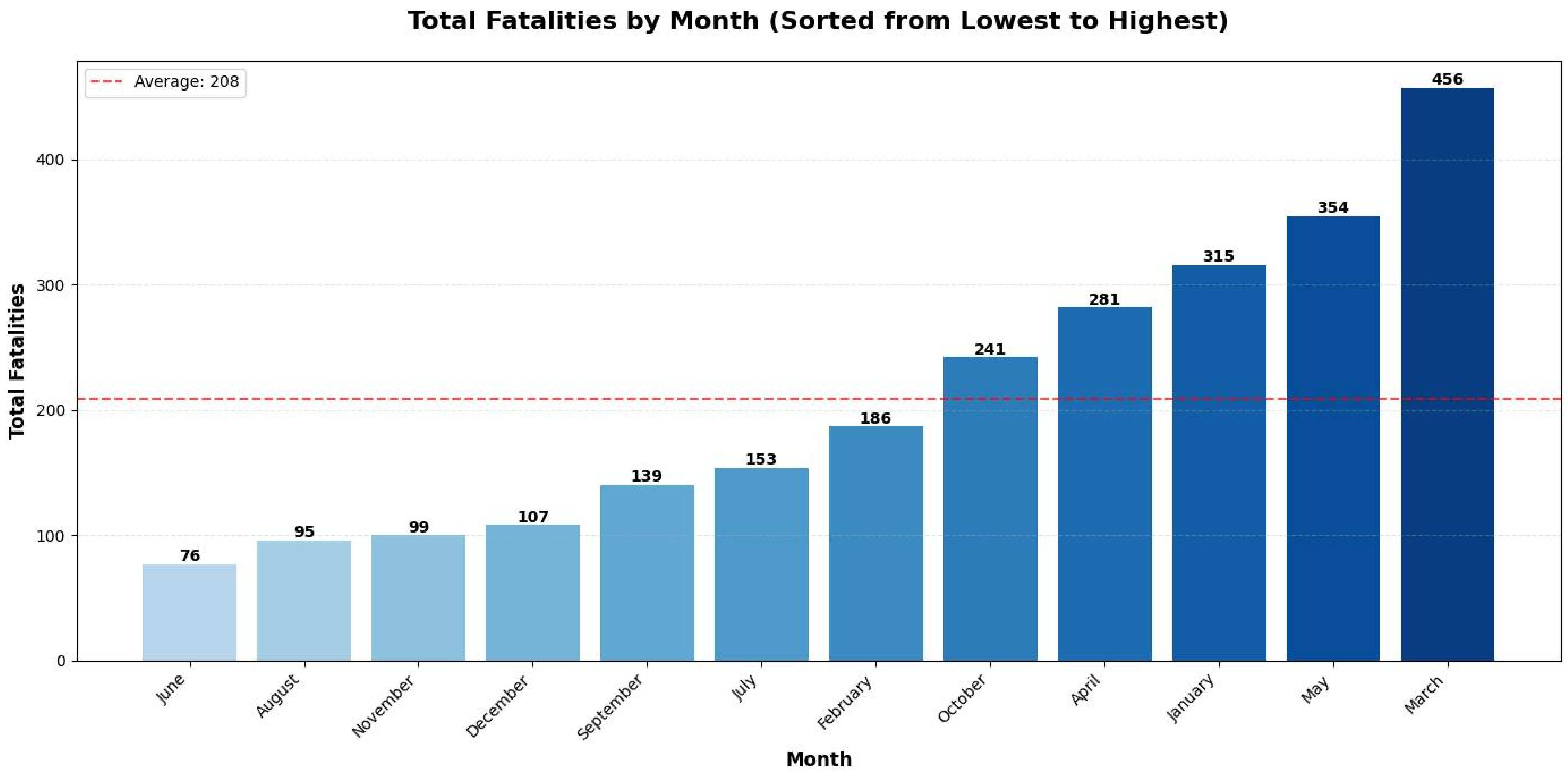
Aircraft_type with lowest and highest fatalities



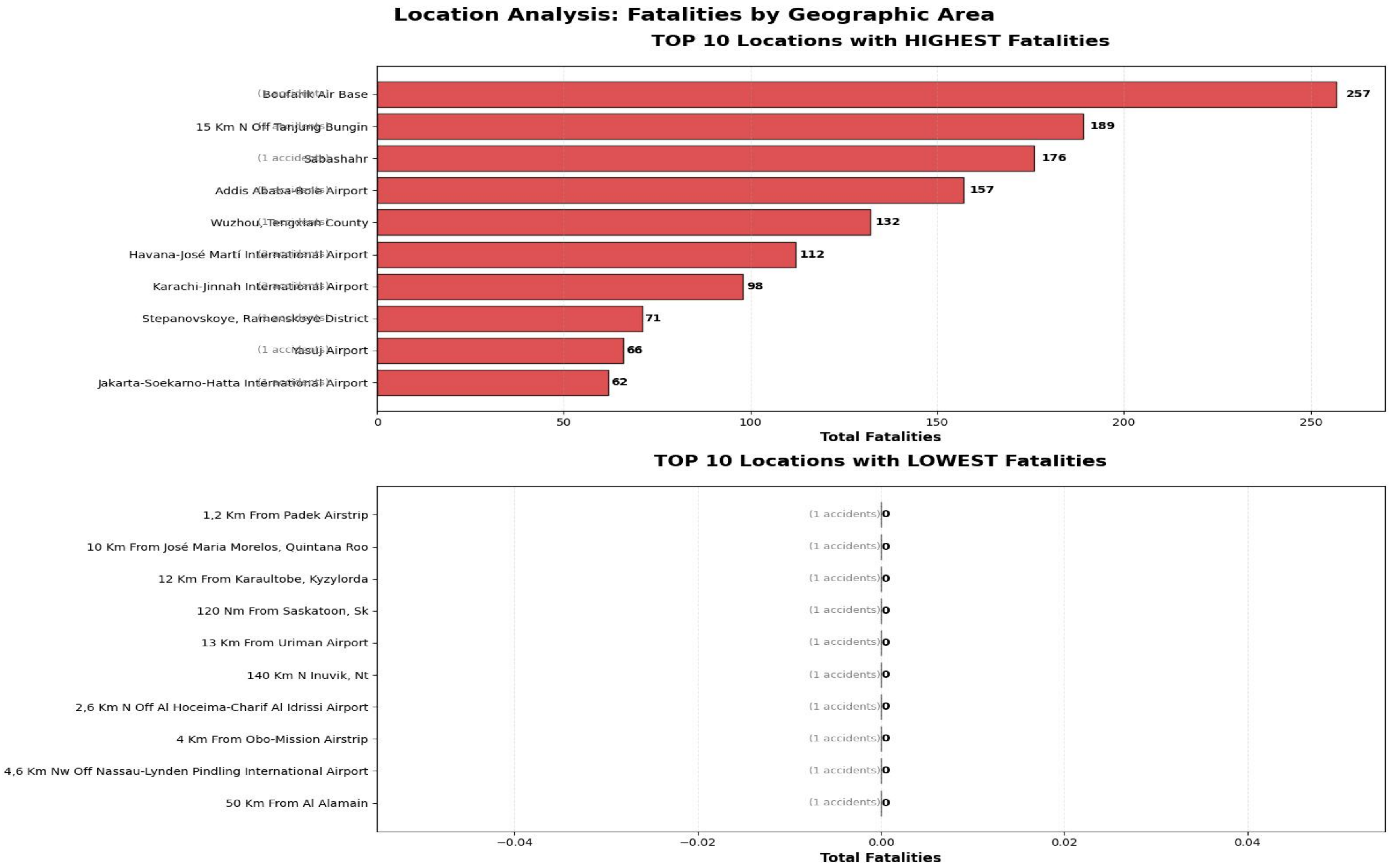
Damage level for aircraft type



Fatalities by the month of the year



Location with lowest and highest fatalities



INSIGHT

The following insight were noted based on:-

Some aircraft types are consistently involved in more severe fatal outcomes than others

Damage level of Aircraft types leads financial loss exposure, long downtime & repair costs respectively, while Mostly minor damage shows survivable accidents

There are relationship between Period of time in a year and fatality cases

Location of aircraft determine fatalities cases

RECOMMENDATION

- Prioritize aircraft types with historically low accident rates

- Improved safety system should be primary factor in investing in aircraft models

- Consider safety history in acquiring aircraft type

- Frequently linked aircraft categories which are prone to severe or fatal incidents should be avoided

Next Steps

To develop an Operational Risk tools for Aircraft type selection and then converting each of the finding into numeric risk scores per objectives for example

Fatality rate per aircraft type, % of destroyed and substatial damage level, fatalities per month of the year, Fatality rate by locations